6. Project Testing Phase

FitBuddy – AI Fitness Plan Generator using Gemini Models

Phase Report

Prepared from the supplied FitBuddy project documentation and organized into a project lifecycle deliverable.

1. Testing Objectives

Verify that user input is validated, workout plans and tips are generated, plans persist in SQLite, feedback updates the plan, and the admin page displays stored data.

2. Functional Test Cases

TC-01: GET / returns the home page. TC-02: valid form input submits successfully. TC-03: generated result includes seven day labels. TC-04: nutrition/recovery tip is displayed. TC-05: saved user and plan appear in the database. TC-06: valid feedback updates the plan. TC-07: unknown user ID produces a clear message. TC-08: admin page lists stored users and plan details.

3. Validation Tests

Try missing username, invalid age, non-positive weight, unsupported intensity, blank goal, blank feedback, and an unknown user ID. The app should reject invalid input or show a helpful message instead of crashing.

4. Integration Tests

Test the browser form through FastAPI, the Gemini API with a valid key, SQLAlchemy persistence, template rendering, and the feedback-to-update flow.

5. API and Local Checks

Start with uvicorn app.main:app --reload. Visit http://127.0.0.1:8000 and http://127.0.0.1:8000/docs. Check terminal logs for errors and confirm static CSS loads.

6. Error Scenarios

Missing API key; network failure; quota/rate-limit response; malformed AI output; database write failure; invalid form submission. Error messages should be understandable and should not reveal secret keys.

7. Test Results Template

For each test, record: test ID, input, expected result, actual result, status (Pass/Fail), date, and notes. Only mark a test as passed after executing it.

8. Testing Outcome

Complete the test checklist after running the application locally. Fix failures and rerun regression tests before the demonstration.

Note: This document describes the intended design and workflow. Test outcomes should be recorded only after the application has been run and verified.